

Protik Bose Pranto

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SUMMARY

I am currently a PhD student in Computer Science at Arizona State University, focusing on geospatial machine learning, urban informatics, and remote sensing to advance urban planning and sustainability. I develop geospatial pipelines leveraging LiDAR, aerial imagery, and street-level vision to map pedestrian infrastructure, inventory urban trees, and evaluate shade and extreme heat exposure. My work blends statistical and geospatial analysis with AI-driven models to turn pixels into actionable insights for planners and communities. I am especially interested in city-scale systems that are explainable, reproducible, and deployable. I also enjoy connecting with peers to drive collaborative research and contribute to community-focused projects.

RESEARCH AREA

Geospatial Machine Learning, Computer Vision, Remote Sensing, Spatial Statistics, Urban Informatics

EDUCATION

Arizona State University Ph.D. in Computer Science	Expected December 2027
Arizona State University M.S. in Computer Science Relevant Coursework: Data Mining, Statistical Machine Learning, ML for Remote Sensing	December 2025
Bangladesh University of Engineering and Technology Bachelor's in Computer Science and Engineering	February 2021

RESEARCH EXPERIENCE

SHaDE Lab: Sensable Heatscapes and Digital Environments Lab Graduate Student Assistant with Dr. Ariane Middel	2023 – Present
Graph Drawing and Information Visualization Laboratory, BUET Undergraduate Researcher with Dr. Md. Saidur Rahman	2019 – 2021

PUBLICATION

- **Protik Bose Pranto**, Isaac Buo, Peter Price, Paige Rearick, Ariane Middel. *Short Paper*: "A Multi-Source LiDAR and Imagery Pipeline for Urban Tree Inventory in Arid Environments". In **Review** at **ACM SIGSPATIAL 2026**.
- **Protik Bose Pranto**, Isaac Buo, Ariane Middel. *Paper*: "WayNet: A Semi-Automated Framework for Mapping Urban Pedestrian Infrastructure Using Street-Level Imagery". In **Review** at **International Journal of Geographical Information Science**.
- **Protik Bose Pranto**, Minhazul Islam, Ripon Saha, Abimelec Mercado Rivera, and Namig Abbasov. *Paper*: "From Hubs to Deserts: Mapping and Explaining Urban Cultural Accessibility". In **Accepted** at **Urban AI (ACM SIGSPATIAL)**, 2025.
- **Protik Bose Pranto**. *Paper*: "Satire or Fake News? Machine Learning-Based Approaches to Resolve the Dilemma". In: **Accepted** at *International Conference on Electrical, Computer, Communications and Mechatronics Engineering, ICECCME (2024)*.
- **Protik Bose Pranto**, Waqar Hassan Khan, Sahar Abdelnabi, Rebecca Weil, Mario Fritz and Rakibul Hasan. *Paper*: "From Bad to Worse: Using Private Data to Propagate Disinformation on Online Platforms with a Greater Efficiency". In: **Accepted** at *Design x Policy, CHI Workshop (2023)*.
- **Protik Bose Pranto**, Bishal Basak Papan, and Md. Saidur Rahman. *Paper*: "k-Safe Labelings of Connected Graphs". In: **Accepted** at *IEEE International Conference on Telecommunications and Photonics, ICTP (2021)*.
- Bishal Basak Papan, **Protik Bose Pranto**, and Md. Saidur Rahman. *Paper*: "On 2-Interval Pairwise Compatibility Properties of Two Classes of Grid Graphs". In: **Accepted** at *The Computer Journal, COMPJ (2021)*.

POSTER/PRESENTATION

- **Protik Bose Pranto.** *Talk:* "Walk in the Shade: Citywide Route Maps". In: **Accepted** at *Three Minute Thesis, 3MT* (2026).
- **Protik Bose Pranto, Isaac Buo, Ariane Middel.** *Poster:* "Scalable LiDAR-Based Framework for Automated Tree Inventory and Management". In: **Accepted** at *CAP LTER All-Scientists Meeting and Poster Symposium, CAP Poster Symposium* (2026).
- **Protik Bose Pranto, Isaac Buo, Ariane Middel.** *Presentation:* "WalkNet: A Vision-Based Framework for Mapping Urban Pedestrian Infrastructure". In: **Accepted** at *American Geophysical Union, AGU* (2025).
- **Protik Bose Pranto, Waqar Hassan Khan, Ariane Middel.** *Poster:* "A Systematic Literature Review on Urban Climate Informatics". In: **Accepted** at *Urban Climate Research Center Poster Event, UCRC* (2025).
- **Protik Bose Pranto, Waqar Hassan Khan, Sahar Abdelnabi, Rebecca Weil, Mario Fritz, and Rakibul Hasan.** *Poster:* "Understanding the Effect of Private Data in Disinformation Propagation". In: **Accepted** at *Symposium on Usable Privacy and Security, SOUPS* (2023).
- **Waqar Hassan Khan, Protik Bose Pranto, Tianyi Yang, Rakibul Hasan.** *Poster:* "Exploring Privacy and Security Concerns of EdTech Users: A Qualitative Analysis of User Written Reviews". In: **Accepted** at *Symposium on Usable Privacy and Security, SOUPS* (2023).
- **Protik Bose Pranto.** *Presentation:* "Targeted Misinformation based on Personal Data". In: **Accepted** at *ASU CyberSecurity Symposium, ASUCSS* (2023).

SELECTED RESEARCH EXPERIENCE

- **Literature Review on Urban Climate Informatics:** Conducting a comprehensive systematic literature review to explore the intersection of urban climate and computer science and examine how this cross-disciplinary approach addresses various urban climate challenges. Our initial findings show a growing interest in social media and crowdsourced approaches in urban climate research.
- **Pedestrian Network Mapping:** Developed a scalable framework to generate detailed pedestrian networks (sidewalks, crosswalks, public access paths), integrating sidewalk width and road type, using vision-based segmentation models, street-level imagery, and open geospatial data. Applied it to Phoenix, Arizona, achieving 78.2% classification accuracy.
- **LiDAR-Based Urban Tree Mapping:** Developed an automated pipeline using LiDAR and ML to create a detailed, georeferenced tree inventory. The process involved classifying LiDAR data with NDVI, building, and land cover data, followed by individual tree segmentation using GDAL polygonization, statistical, and clustering methods (86% accuracy). Morphological metrics were calculated for each tree, and using U-Net, trees were classified into mesic and xeric types.
- **Cultural Accessibility Analysis:** Integrated open city data, socio-economic datasets, and cultural infrastructure information to map cultural hubs and deserts across New York City. Developed a novel "Socio-Cultural Accessibility Score" using statistical analysis, machine learning (R^2 of 0.91), a weight-decaying algorithm, and an interpretable model (SHAP) to identify and explain underserved and overserved regions from cultural, economic, and health perspectives.
- **Fake News and Misinformation Detection:** Developed and evaluated ML models, including traditional algorithms and transformer-based models (e.g., BERT, XLM-RoBERTa), for fake and misleading news detection across various domains. Applied data preprocessing techniques to optimize model performance, with SVM achieving the best results on small datasets (86% accuracy). Used novel text augmentation strategies to enhance model accuracy, achieving 97% accuracy with XLM-RoBERTa. Additionally, applied BERT for misinformation detection in a regional language (Bengali), obtaining a 97% F1-score.

DATASET CONTRIBUTION

- Isaac Buo, Waqar Hassan Khan, **Protik Bose Pranto**, and Ariane Middel. *"Hourly Mean Radiant Temperature Distribution on a Summer Day, Maricopa County, Arizona, USA"*. DOI: [10.6073/pasta/300ddb6879bab8a5e0208f6df3ed2cc0](https://doi.org/10.6073/pasta/300ddb6879bab8a5e0208f6df3ed2cc0)

GRANTS

- **CAP LTER Grad Grant:** Awarded \$5000 in funding from the Central Arizona–Phoenix Long-Term Ecological Research program (CAP LTER), a National Science Foundation (NSF)-funded initiative focusing on urban ecology. As the lead researcher on the project "Mapping the Distribution and Characteristics of Native and Non-Native Tree Species," This work supports urban forestry planning and heat mitigation strategies in the Phoenix-Tempe region.
- **Travel Grant:** SCAI conference funding (\$250), Graduate College Travel Award (\$300), GSG Travel Fund (\$700)
- **USENIX Security Student Grant:** Awarded \$700 to attend the USENIX Security'23 conference, where I presented my work on misinformation. This grant supported my participation in the conference.

ACADEMIC SERVICES

- **Peer Reviewer for Papers** ML4RS 2026, ICECET 2026, ICECCME 2025, ICECET 2025, ACDSA 2025
- **Conference Session Chair** ICECCME 2024
- **Invited Talk** ASU CyberSecurity Symposium 2023
- **Student Volunteer** AIES 2025, SOUPS 2023

ACHIEVEMENTS

- Finalist at **Three Minute Thesis (3MT)** 2026
- Secured runners-up at **SpaceHACK for Sustainability Hackathon** 2025
- Won 3rd place in the graduate category at the **UCRC Poster Event** 2025
- Awarded **USENIX Security Student Grant** 2023
- Got accepted into the **post-CHI summer school** on Usable Privacy and Security 2023
- Been awarded the **SCAI Doctoral Fellowship** 2022
- Completed **Google Foobar Challenge** 2020
- Ranked 3rd place in **South Asia Center for Media in Development** 2020
- Winner of the **HackTheCode** contest in **Google Cloud DevFest** 2019

LEADERSHIP EXPERIENCES

- **Coordinator**, OLsA-Arizona Laboratorians Chapter 2024 - Present
- **Student Representative**, Bangladesh Student Association, Arizona State University 2023 - Present
- **Organising Member**, Laboratorian Association of BUET 2019 - 2020

INDUSTRY EXPERIENCE

GIS Tech Research Intern May 2026 – Aug 2026
DLC Resources
Developing geospatial computer vision pipelines for tree inventory mapping and land use classification using drone imagery, street-view data, GIS, remote sensing, and machine learning.

Consultant Mar 2022 – Jul 2022
Start Network
Contributed to the Forecast-based Warning, Analysis, and Response Network (**FOREWARN**) project by developing data science workflows, applying advanced analytics, and providing technical expertise to support a peer-reviewed publication.

Software Engineer Mar 2021 – Feb 2022
Chaldal Engineering
An online grocery and food delivery platform. Enhanced customer-facing platforms (website, mobile app, APIs, search catalog) and optimized order management systems. Contributed to the development of the **EggShell** frontend framework, integrating F#, React, Fable, and ReactXP, enabling unified and faster development for both web and Android platforms.

TEACHING ASSISTANT

- **CSE 100 (Principle of Programming with C++)**
The course focuses on programming concepts, problem-solving, and program design.
- **CSE 360 (Introduction to Software Engineering)**
Introduces software engineering principles such as life cycle models, development methods, UML, project management, testing, and quality assurance standards.
- **CSE 467 (Data and Information Security)**
This course is intended to provide students with an introductory understanding of the technical and behavioral mechanisms for information security and privacy.
- **CSE 470 (Computer Graphics)**
This course introduces the basic concepts of interactive computer graphics, realistic rendering, and 3D viewing.
- **CSE 477 (Intro Computer-Aided Geometric Design)**
The course introduces basic concepts of 3-D computer geometry, including curves, surfaces, and meshes.

- **CSE 485 (ASU Capstone Project)**

It is a project-based course linking students with industry and faculty-sponsored projects, cultivating practical experience, and preparing them for the professional world.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, C++, Java, Dart, Typescript, F#, Flutter.
- **ML & Data Science:** Keras, PyTorch, Scikit-learn, Statistics, Time-series analysis, NLP, Feature extraction, SHAP, Image segmentation, Regression/Classification, Reproducibility (Weights & Biases), Data visualization.
- **Geospatial & Computer Vision:** OpenCV, CUDA, Geemap, GDAL, Rasterio, GeoPandas, Shapely, PyProj, White-Box Geospatial, Spatial analysis, LiDAR analysis.
- **Tools & Platforms:** Docker, Google Earth Engine, QGIS, Git, Docker, Jupyter, LAStools, Cyberduck, Excel.

SELECTED ACADEMIC PROJECTS

Vasha-Sikkha

Tech Stack: Dart, Flutter, Firebase

Developed a mobile application for interactive English learning with speaking, reading, listening, and writing modules. Implemented score tracking, progress monitoring, and a real-time leaderboard using a Firebase backend.

CovidLife

Tech Stack: Dart, Flutter, REST APIs

Built a health monitoring application displaying COVID-19 data for Bangladesh. Integrated two REST APIs for district-level and daily statistics, and added appointment scheduling, in-app calling, and messaging features.

Real-Time Vehicle Detection

Tech Stack: Python, YOLOv5, OpenCV

Developed an object detection pipeline for fisheye traffic camera footage using YOLOv5. Applied geometric transformations for distortion correction, improving multi-vehicle tracking accuracy.

Gesture Sensed Snake Game

Tech Stack: C, C++, Embedded Systems, Atmega32, Accelerometer

Designed an embedded motion-controlled game using an Atmega32 microcontroller and accelerometer. Controlled 8x8 RGB matrices through decoder circuits for real-time rendering.

REFERENCES

Dr. Ariane Middel

Associate Professor, School of Computing and Augmented Intelligence, Arizona State University

Relationship: Ph.D. Advisor

Email: amiddel@asu.edu

Dr. Isaac Buo

Postdoctoral Scholar, School of Arts, Media and Engineering, Arizona State University

Relationship: Research Collaborator

Email: ibuol@asu.edu

Dr. Nishan Kumar Biswas

Associate Scientist, Hydrological Research Laboratory, NASA Goddard Space Flight Center

Relationship: Research Collaborator

Email: n.biswas@nasa.gov

Dr. Lynn Robert Carter

Professor of Practice, School of Computing and Augmented Intelligence, Arizona State University

Relationship: Teaching Mentor

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